Question Summary

- What do you mean by radix?
- Solve the following number conversion-

- $(10010011)_2=(?)_{16}=(?)_8=(?)_{10}$
- $(10.011)_2=(?)_8=(?)_{16}=(?)_{10}$
- $(365.34)_8=(?)_{16}=(?)_2=(?)_{10}$
- $(572.6)_{10}=(?)_2=(?)_8=(?)_{16}$
- $(6CD.4E)_{16}=(?)_{10}=(?)_8=(?)_2$

Digital Electronics

Boolean Algebra

De Morgan's Theorem

Boolean algebra

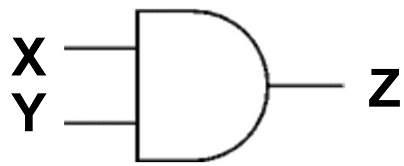
- A Boolean algebra consist of ...
  - A set of elements B
  - Binary operators {+ , •} Boolean sum and product
  - A unary operation { ' } (or {¯}) example: A′ or $\bar{A}$

  - AND for • Boolean Product.
  - OR for + Boolean Sum.
  - NOT for ` Complement.

- Boolean algebra and the following axioms
  - 1. The set B contains at least two elements {a b} with $a \neq b$
  - 2. Commutative: $a+b = b+a$ $a \bullet b = b \bullet a$
  - 3. Associative: $a+(b+c) = (a+b)+c$ $a \bullet (b \bullet c) = (a \bullet b) \bullet c$
  - 4. Identity: $a+0 = a$ $a \bullet 1 = a$
  - 5. Distributive: $a+(b \bullet c)=(a+b) \bullet (a+c)$ $a \bullet (b+c)=(a \bullet b)+(a \bullet c)$
  - 6. Complementarity: $a+a' = 1$ $a \bullet a' = 0$

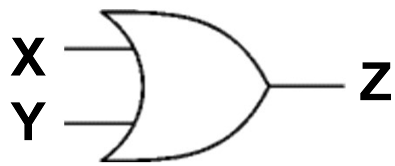
Logic Gates (AND, OR, Not) & Truth Table

- AND $X \bullet Y$ XY



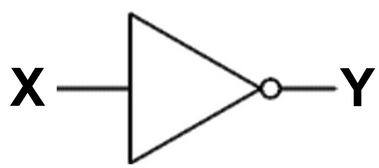
X	Y	Z
0	0	0
0	1	0
1	0	0
1	1	1

- OR $X+Y$



X	Y	Z
0	0	0
0	1	1
1	0	1
1	1	1

- NOT $\overline{X}$ X'



X	Y
0	1
1	0

Logic functions and Boolean algebra

- *Any* logic function that is expressible as a truth table can be written in Boolean algebra using +, •, and '

X	Y	Z
0	0	0
0	1	0
1	0	0
1	1	1

Z=X•Y

X	Y	X'	Z
0	0	1	0
0	1	1	1
1	0	0	0
1	1	0	0

Z=X'•Y

X	Y	X'	Y'	X • Y	X' •Y'	Z
0	0	1	1	0	1	1
0	1	1	0	0	0	0
1	0	0	1	0	0	0
1	1	0	0	1	0	1

Z=(X•Y)+(X' •Y')

Some notation

- Priorities: $\overline{A} \bullet B + C = ((\overline{A}) \bullet B) + C$
- Variables and their complements are sometimes called literals

Useful laws and theorems

Identity:	$X + 0 = X$	Dual: $X \bullet 1 = X$
Null:	$X + 1 = 1$	Dual: $X \bullet 0 = 0$
Idempotent:	$X + X = X$	Dual: $X \bullet X = X$
Involution:	$(X')' = X$	
Complementarity:	$X + X' = 1$	Dual: $X \bullet X' = 0$
Commutative:	$X + Y = Y + X$	Dual: $X \bullet Y = Y \bullet X$
Associative:	$(X+Y)+Z=X+(Y+Z)$	Dual: $(X\bullet Y)\bullet Z=X\bullet(Y\bullet Z)$
Distributive:	$X\bullet(Y+Z)=(X\bullet Y)+(X\bullet Z)$	Dual: $X+(Y\bullet Z)=(X+Y)\bullet(X+Z)$
Uniting:	$X\bullet Y+X\bullet Y'=X$	Dual: $(X+Y)\bullet(X+Y')=X$

Useful laws and theorems (con't)

Absorption: $X+X \bullet Y=X$ Dual: $X \bullet (X+Y)=X$

Absorption (#2): $(X+Y') \bullet Y=X \bullet Y$ Dual: $(X \bullet Y')+Y=X+Y$

de Morgan's: $(X+Y+...)'=X' \bullet Y' \bullet ...$ Dual: $(X \bullet Y \bullet ...)'=X'+Y'+...$

Duality: $(X+Y+...)^D=X \bullet Y \bullet ...$ Dual: $(X \bullet Y \bullet ...)^D=X+Y+...$

Multiplying & factoring: $(X+Y) \bullet (X'+Z)=X \bullet Z+X' \bullet Y$

Dual: $X \bullet Y+X' \bullet Z=(X+Z) \bullet (X'+Y)$

Consensus: $(X \bullet Y)+(Y \bullet Z)+(X' \bullet Z)= X \bullet Y+X' \bullet Z$

Dual: $(X+Y) \bullet (Y+Z) \bullet (X'+Z)=(X+Y) \bullet (X'+Z)$

Truth Table and Boolean Function

SOP – Sum-of-Product Representation

Minterms: Minterms are a way to represent and describe specific combinations of input variables in a Boolean function.

They are also known as standard products or canonical products.

Minterms are used to simplify and analyze Boolean expressions.

Truth Table and Boolean Function

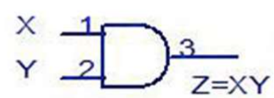
SOP – Sum-of-Product Representation

Minterms: Minterms are a way to represent and describe specific combinations of input variables in a Boolean function.

Index	x	y	Minterm
0	0	0	$m_0 = \overline{x}\,\overline{y}$
1	0	1	$m_1 = \overline{x}\,y$
2	1	0	$m_2 = x\,\overline{y}$
3	1	1	$m_3 = x\,y$

AND Gate

INPUTS		OUTPUT
X	Y	Z
0	0	0
0	1	0
1	0	0
1	1	1



OR Gate

INPUTS		OUTPUT
X	Y	Z
0	0	0
0	1	1
1	0	1
1	1	1

$$
\begin{aligned}
z &= X'Y + XY' + XY \\
&= X'Y + X(Y' + Y) \\
&= X'Y + X \\
&= (X' + X)(Y + X) \\
&= X + Y
\end{aligned}
$$

Truth Table and Boolean Function

SOP – Sum-of-Product Representation

Minterms: Minterms are a way to represent and describe specific combinations of input variables in a Boolean function.

				Minterms
X	Y	Z		Product Terms
0	0	0		$m_0 = \overline{X} \cdot \overline{Y} \cdot \overline{Z} = \min(\overline{X}, \overline{Y}, \overline{Z})$
0	0	1		$m_1 = \overline{X} \cdot \overline{Y} \cdot Z = \min(\overline{X}, \overline{Y}, Z)$
0	1	0		$m_2 = \overline{X} \cdot Y \cdot \overline{Z} = \min(\overline{X}, Y, \overline{Z})$
0	1	1		$m_3 = \overline{X} \cdot Y \cdot Z = \min(\overline{X}, Y, Z)$
1	0	0		$m_4 = X \cdot \overline{Y} \cdot \overline{Z} = \min(X, \overline{Y}, \overline{Z})$
1	0	1		$m_5 = X \cdot \overline{Y} \cdot Z = \min(X, \overline{Y}, Z)$
1	1	0		$m_6 = X \cdot Y \cdot \overline{Z} = \min(X, Y, \overline{Z})$
1	1	1		$m_7 = X \cdot Y \cdot Z = \min(X, Y, Z)$

Two key concepts

- de Morgan's Theorem
  - Procedure for complementing Boolean functions
  - Replace: • with +, + with •, 0 with 1, and 1 with 0
  - Replace all variables with their complements

DeMorgan’s Theorem

DeMorgan’s First Theorem

DeMorgan’s First theorem proves that when two (or more) input variables are AND’ed and negated, they are equivalent to the OR of the complements of the individual variables. Thus the equivalent of the NAND function and is a negative-OR function proving that (A.B)’ = A’+B’ and we can show this using the following table.

Verifying DeMorgan’s First Theorem using Truth Table

Inputs		Truth Table Outputs For Each Term					
B	A	(A.B)	(A.B)′	A′	B′	A′ + B′	
0	0	0	**1**	1	1	**1**	
0	1	0	**1**	0	1	**1**	
1	0	0	**1**	1	0	**1**	
1	1	1	**0**	0	0	**0**	

We can also show that (A.B)’ = A’+B’ using logic gates as shown.

DeMorgan's Theorem (Cont..)

DeMorgan's Second Theorem

DeMorgan's Second theorem proves that when two (or more) input variables are OR'ed and negated, they are equivalent to the AND of the complements of the individual variables. Thus the equivalent of the NOR function and is a negative-AND function proving that (A+B)' = A'.B' and again we can show this using the following truth table.

Verifying DeMorgan's Second Theorem using Truth Table

Inputs		Truth Table Outputs For Each Term					
B	A	A+B	(A+B)′	A′	B′	A′ . B′	
0	0	0	**1**	1	1	**1**	
0	1	1	**0**	0	1	**0**	
1	0	1	**0**	1	0	**0**	
1	1	1	**0**	0	0	**0**	

We can also show that (A+B)' = A'.B' using logic gates as shown.

Proving theorems

- Example 1: Prove the uniting theorem-- $X \bullet Y + X \bullet Y' = X$

Distributive	$X \bullet Y + X \bullet Y' = X \bullet (Y + Y')$
Complementarity	$= X \bullet (1)$
Identity	$= X$

- Example 2: Prove the absorption theorem-- $X + X \bullet Y = X$

Identity	$X + X \bullet Y = (X \bullet 1) + (X \bullet Y)$
Distributive	$= X \bullet (1 + Y)$
Null	$= X \bullet (1)$
Identity	$= X$

Proving theorems

- Example 3: Prove the consensus theorem--
  $(XY)+(YZ)+(X'Z)= XY+X'Z$

Proving theorems

- Example 3: Prove the consensus theorem-- $(XY)+(YZ)+(X'Z)= XY+X'Z$

Complementarity	$XY+YZ+X'Z$	$= XY+(X+X')YZ + X'Z$
Distributive		$= XYZ+XY+X'YZ+X'Z$

- *Use absorption {AB+A=A} with A=XY and B=Z*

		$= XY+X'YZ+X'Z$
Rearrange terms		$= XY+X'ZY+X'Z$

- *Use absorption {AB+A=A} with A=X'Z and B=Y*

$$XY+YZ+X'Z = XY+X'Z$$

de Morgan's Theorem

- Use de Morgan's Theorem to find complements
- Example: $F=(A+B)\bullet(A'+C)$, so $F'=(A'\bullet B')+(A\bullet C')$

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

A	B	C	F'
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

One more example of logic simplification

- Example:

$$Z = A'BC + AB'C' + AB'C + ABC' + ABC$$

One more example of logic simplification

- Example:

$$Z = A'BC + AB'C' + AB'C + ABC' + ABC$$

$$= A'BC + AB'(C' + C) + AB(C' + C) \quad \text{distributive}$$
$$= A'BC + AB' + AB \quad \text{complementary}$$
$$= A'BC + A(B' + B) \quad \text{distributive}$$
$$= A'BC + A \quad \text{complementary}$$

$$= BC + A \quad \text{absorption \#2 Duality}$$

$(X \bullet Y') + Y = X + Y$ with $X = BC$ and $Y = A$

Few Problems

Simplify following Boolean function.

- $Z = AB + BC + B'C = AB + C$
- $Z = A + A'B = A + B$
- $Z = A'B'C + A'BC + AB' = A'C + AB'$
- $Z = AB + (AC)' + AB'C(AB + C) = 1$
- $Z = AB(B'C + AC) = ABC$
- $Z = (A + B)(A + C) = A + BC$
- $Z = (A + B' + C')(A + B' + C)(A + B + C') = A + B'C'$

Few Problems

Minimize the following Boolean equation and draw the circuit diagram using logic gate.

i) $Z= (A+B)(A+C')(B'+C')=AB'C+A'BC'$

ii) $Z= (X+Y'+XY)(X+Y)(XY)$

iii) $Z= ABC+ABC'+AB'C+A'BC$

iv) $Z=AB+ABCD+ABC+AD$

v) $Z=(AB'+A'B)'AB$

vi) $Z=AB+C(AB'+A'B)$

Few Problems

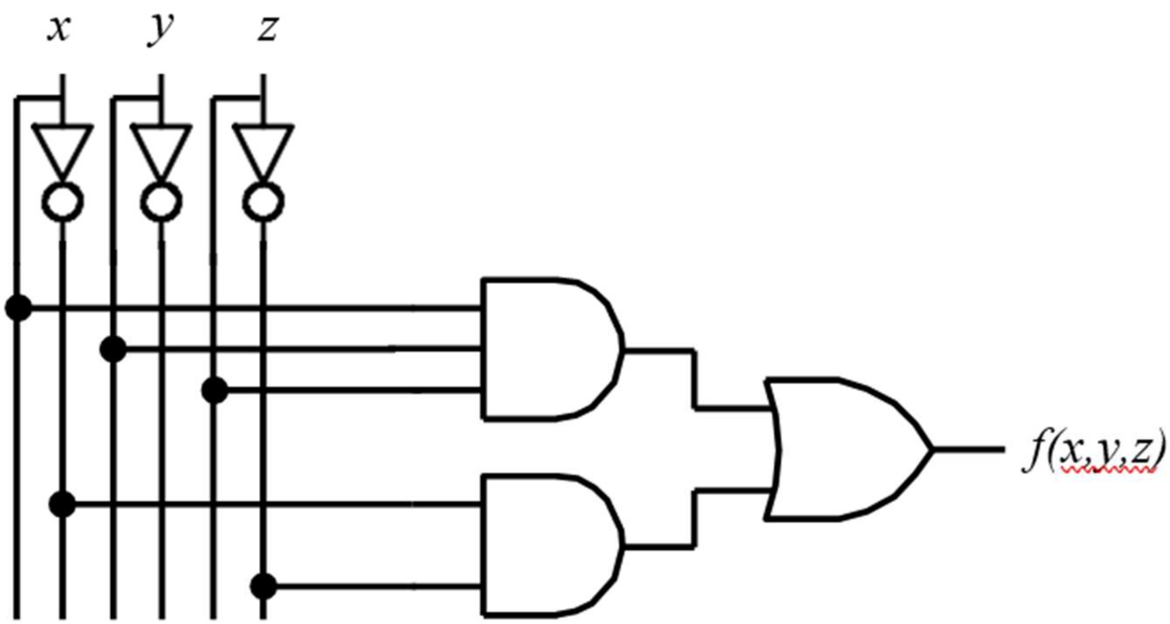
Determine the Boolean expression from the given digital circuits.

i) $Z = (A+B)(A+C')(B'+C') = AB'C + A'BC'$

Few Problems

Determine the Boolean expression from the given digital circuits.

i) Z= (A+B)(A+C′)(B′+C′)=AB′C+A′BC′



Few Problems

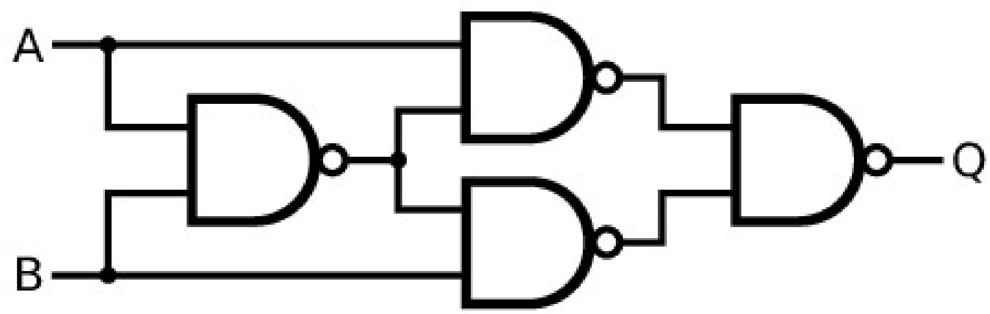
Determine the Boolean expression from the given digital circuits.

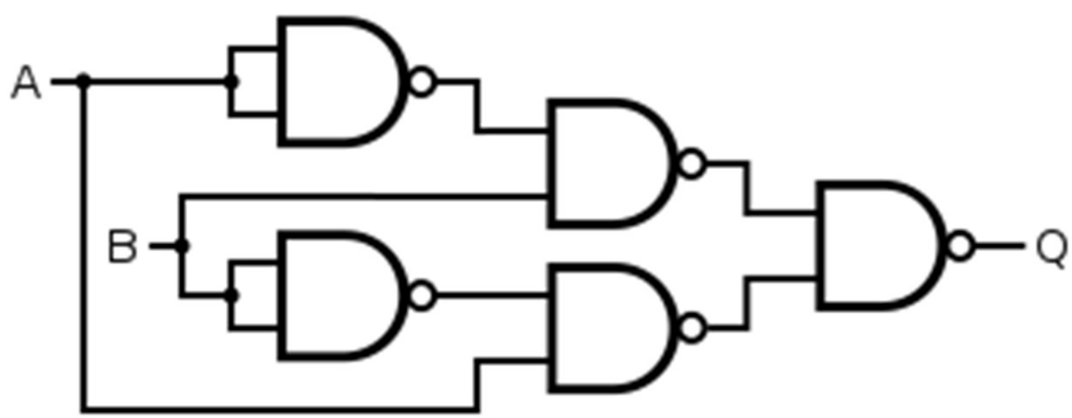
ii) $Z= (X+Y'+XY)(X+Y)(XY)$

Few Problems

Determine the Boolean expression from the given digital circuits.

ii) Z= (X+Y′+XY)(X+Y)(XY)





Few Problems

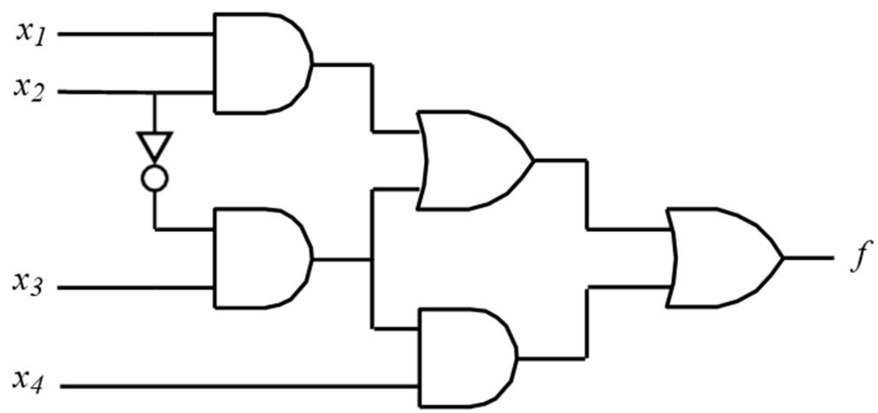
Determine the Boolean expression from the given digital circuits.

iii) $Z = ABC + ABC' + AB'C + A'BC$

Few Problems

Determine the Boolean expression from the given digital circuits.

iii) Z= ABC+ABC′+AB′C+A′BC



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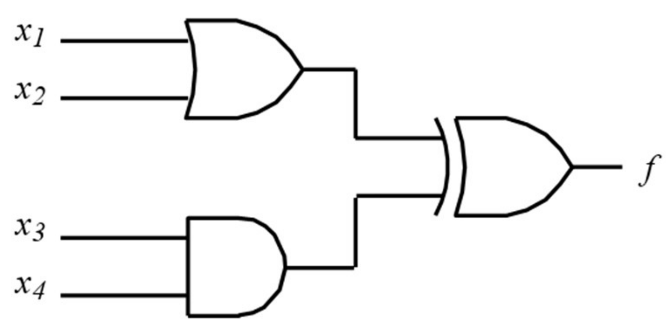
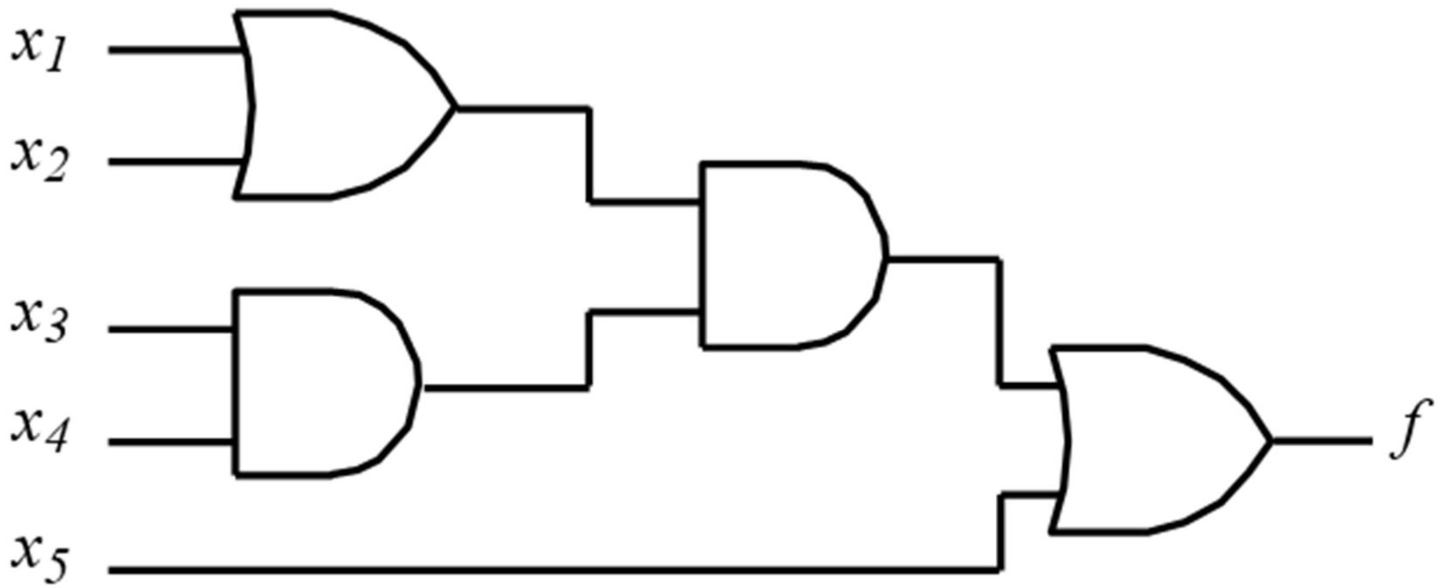


(c)

Few Problems

Draw the equivalent logic circuits of the following Figure using NAND gates only.

i)

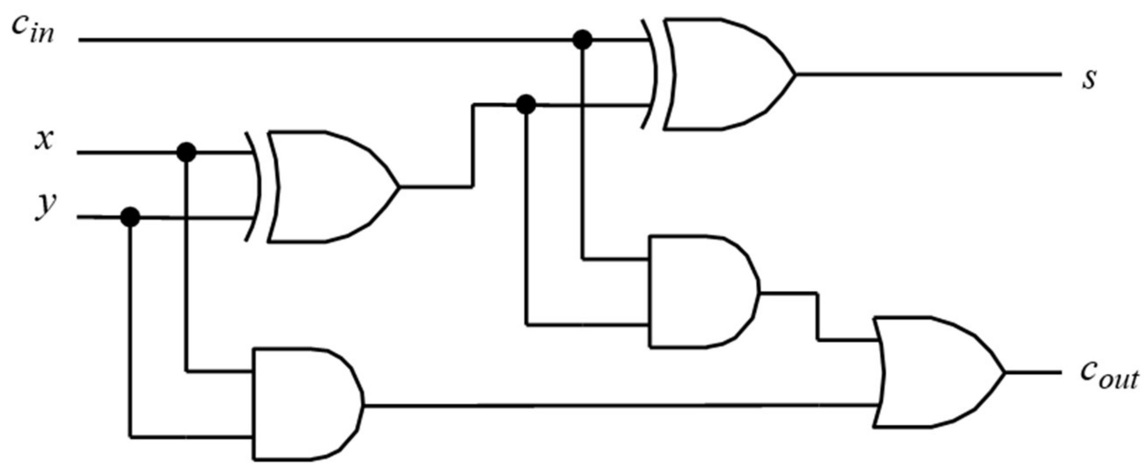


(a)

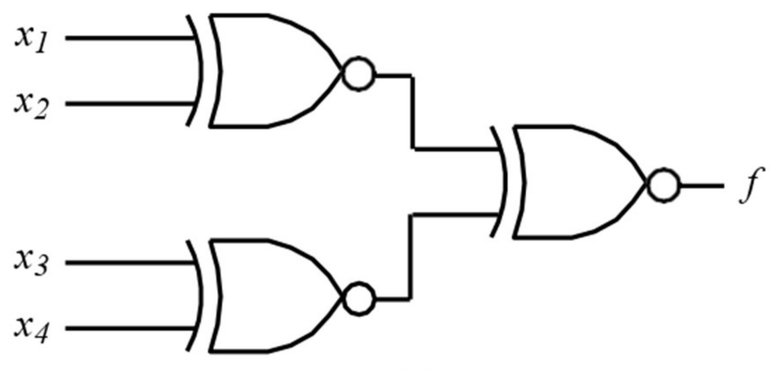
Few Problems

Draw the equivalent logic circuits of the following Figure using NAND gates only.

ii)



(d)

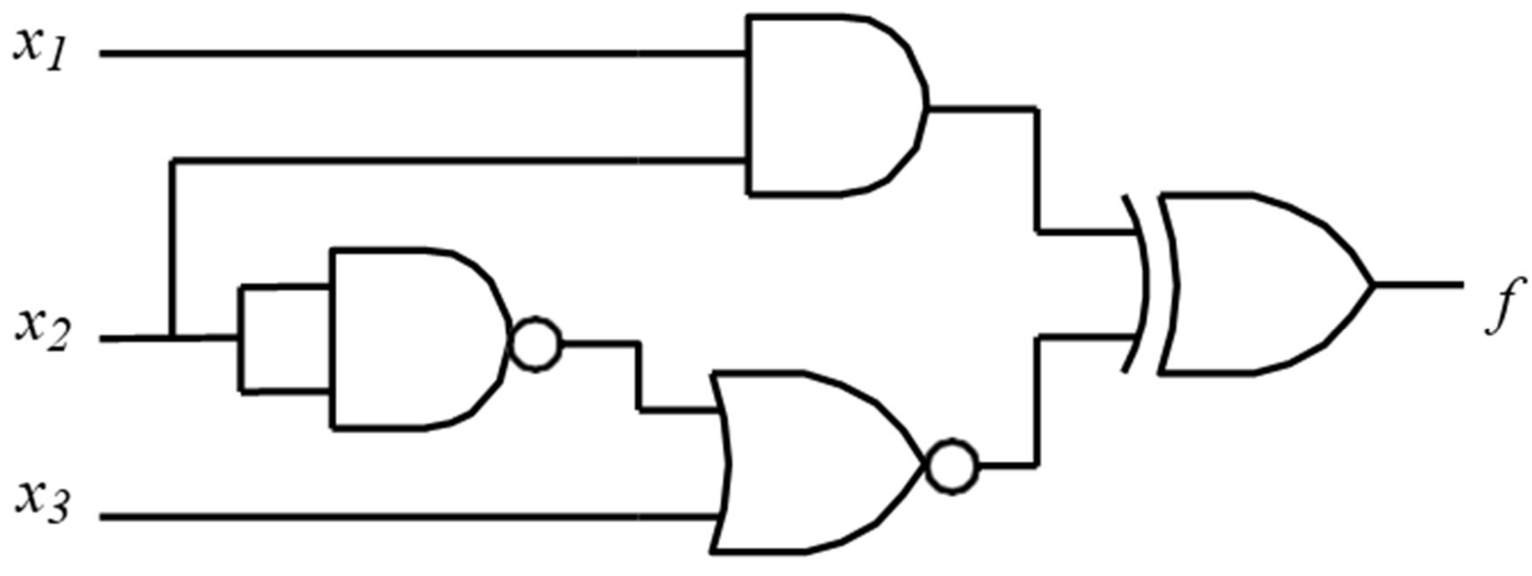


(e)

Few Problems

Draw the equivalent logic circuits of the following Figure using NAND gates only.

iii)



(b)

Thank you